ModelIt!

Teacher's Guide

CRISPR Gene Drive: Should We Eliminate Malaria?

Systems Biology Modeling of Gene Drive Ecology and Ethics

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS3-2, HS-LS4-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS3-2, HS-LS4-5

Make and defend a claim based on evidence that inheritable genetic variations may result from new genetic combinations through meiosis, viable errors occurring during replication, or mutations caused by environmental factors; evaluate the evidence supporting claims that changes in environmental conditions may result in increases in the number of individuals of some species, the emergence of new species over time, and the extinction of other species.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students evaluate competing claims about gene drive deployment using systems biology model data, ecological evidence, and ethical reasoning to construct and defend evidence-based positions.
Disciplinary Core Idea	**LS3.B Variation of Traits / LS4.C Adaptation** — Genetic variation arises through mutation and is shaped by natural selection; gene drives exploit and override natural inheritance mechanisms, with evolutionary consequences for both target and non-target species.
Crosscutting Concept	**Cause and Effect / Systems and System Models** — Students trace causal chains from molecular CRISPR mechanisms through population genetics to ecosystem-level consequences, modeling the system across multiple scales and timescales.

CAST Testing Alignment

MULTIPLE CHOICE:

A gene drive with 95% efficiency is released in a mosquito population. After 15 generations, Wild-Type Gene Frequency begins to increase instead of decrease. Based on your model, what is the most likely explanation?

CONSTRUCTED RESPONSE:

Using your systems biology model, construct an argument for OR against deploying a CRISPR gene drive to eliminate malaria. You must address at least two benefits, two risks, and the role of uncertainty in your recommendation. Reference specific model components and simulation results.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to systems biology and synthetic design in the context of gene drive technology — arguably the most powerful and controversial application of CRISPR genetic engineering. Biotech skill focus: Systems biology and synthetic design. Gene drives have the theoretical capacity to alter or eliminate entire wild species, forcing us to confront questions at the intersection of molecular biology, ecology, evolution, and ethics.

CRISPR Gene Drive Mechanism

A CRISPR gene drive works by encoding the Cas9 protein and a guide RNA within the modified gene itself. When a gene-drive-carrying mosquito mates with a wild-type mosquito, the Cas9 protein cuts the wild-type copy of the gene on the partner chromosome. The cell's DNA repair machinery then uses the gene drive copy as a template, converting the heterozygous cell to homozygous for the drive. This means that instead of inheriting the modified gene 50% of the time (Mendelian), offspring inherit it 99%+ of the time. Over multiple generations, the drive spreads exponentially through the population. Two strategies exist: population suppression drives (carrying genes that reduce female fertility, crashing the population) and population modification drives (carrying genes that make mosquitoes unable to transmit the malaria parasite while maintaining population size).

Malaria: The Scale of the Problem

Malaria kills approximately 600,000 people per year — a child dies every minute. Over 95% of malaria deaths occur in sub-Saharan Africa, and 80% of victims are children under 5. The disease is caused by Plasmodium parasites transmitted by female Anopheles mosquitoes. Despite billions of dollars invested in bed nets, insecticides, and antimalarial drugs, malaria has proven extraordinarily difficult to control because the parasite has evolved resistance to every drug we've developed, and the mosquito has evolved resistance to every insecticide. Gene drives offer a fundamentally different approach — instead of trying to outpace evolution with new chemicals, they co-opt evolution itself to spread the desired trait through the vector population.

Ecological Risk Assessment

The ecological consequences of driving Anopheles mosquitoes extinct are genuinely uncertain. Mosquitoes serve as food for bats, birds, dragonflies, frogs, fish, and spiders. Mosquito larvae are a significant food source in freshwater ecosystems. Some mosquito species pollinate plants (primarily in Arctic regions). However, ecologists are divided on whether mosquitoes are a 'keystone species' whose removal would trigger cascading extinctions, or a replaceable food source whose ecological niche would be filled by other insects. The honest scientific answer is: we don't know. Models can simulate potential cascades, but the complexity of real ecosystems means our predictions carry substantial uncertainty — and the consequences of being wrong are irreversible.

The Gene Drive Governance Challenge

Gene drives present an unprecedented governance challenge because they are inherently transboundary — mosquitoes don't respect national borders. A gene drive released in one country will inevitably spread to neighboring countries and potentially the entire species range. This means the decision to deploy affects every nation in the species' range, yet there is no international body with the authority to make or enforce such decisions. The Convention on Biological Diversity has debated gene drive moratoriums; the African Union has studied the issue extensively; the Target Malaria consortium conducts community engagement in Burkina Faso, Mali, and Uganda. The core ethical tension: children are dying NOW from malaria, but deploying an irreversible technology based on uncertain ecological models risks unforeseen consequences that could last forever. Who decides — and on what evidence basis?

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine components spanning molecular genetics (Gene Drive Efficiency, Wild-Type Gene Frequency), population biology (Mosquito Population, Resistance Evolution, Geographic Spread Rate), human health (Malaria Transmission Rate, Human Malaria Cases), and ecology (Ecological Dependence, Ecosystem Disruption Risk).

E **Establish Relationships**

Students determine that Gene Drive Efficiency drives the decline in Wild-Type Gene Frequency, which suppresses Mosquito Population, reducing Malaria Transmission Rate and Human Malaria Cases — while simultaneously increasing Ecosystem Disruption Risk through Ecological Dependence, with Resistance Evolution working to counteract the drive.

V **Visualize & Model**

Students build a systems biology model connecting molecular gene drive mechanics to population-level dynamics to ecosystem-scale consequences, visualizing the multi-generational spread and its cascading effects.

E **Evaluate Outcomes**

Students run moderate, high-resistance, and maximum-efficiency scenarios to map the outcome space — identifying parameter regions where malaria is eliminated, where resistance halts the drive, and where ecological cascades become severe.

R **Revise & Extend**

Students add Alternative Vector Species, Gene Drive Reversal Mechanism, or Climate Change Interaction to explore more realistic and nuanced scenarios.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"A child dies of malaria every 60 seconds. We have the technology to stop it — by engineering mosquitoes to carry a self-spreading genetic modification that could wipe them out. But there's a catch: once we release it, we can never take it back."*

Watch For

Start a 60-second timer. Tell students that by the time it ends, another child will have died from malaria. Let the silence speak.

Slide 2: Learning Objectives

3 min

Say: *"Today you're going to model a technology that can rewrite the genetics of an entire wild species. And then you're going to decide whether we should use it."*

Watch For

Pre-teach gene drive and CRISPR-Cas9. Show a simple animation of gene drive inheritance versus normal Mendelian inheritance.

Slide 3: Big Question

3 min

Say: *"If you could press a button that saves 600,000 lives a year but might permanently alter ecosystems in ways we can't predict — would you press it?"*

Watch For

Anonymous poll: Press or don't press? Record the class results. Tell them the model will help them make a more informed decision.

Slide 4: LEVER Framework

2 min

Say: *"We're building a systems biology model that connects a molecular event — CRISPR cutting DNA — to the extinction of a species and the potential collapse of ecosystems. That's four scales in one model."*

Watch For

Preview LEVER steps. Map the four scales: molecular, population, human health, ecosystem. Emphasize that systems biology integrates all of them.

Slide 5: Activity 1: Components

8 min

Say: *"Only ONE of these nine components is truly under human control — the efficiency of the gene drive we engineer. Everything else is biology responding to our intervention. What does that tell you about the risks?"*

Watch For

Guide component sorting. Discuss why Gene Drive Efficiency is the only truly controllable variable. Map each component to its biological scale.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the cascade: A molecular edit in a mosquito embryo leads to the potential extinction of a species and the collapse of food webs. Map every connection."*

Watch For

Students trace the full causal chain from Gene Drive Efficiency through population suppression to ecological cascade. Identify where Resistance Evolution acts as a brake on the system.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run these three scenarios and find out: Is there a gene drive setting that eliminates malaria without devastating the ecosystem? Or is that impossible?"*

Watch For

Students predict outcomes for all three scenarios, then run simulations. Track Human Malaria Cases and Ecosystem Disruption Risk simultaneously to visualize the trade-off.

Slide 8: Discoveries

5 min

Say: *"Your model just showed you what the world's top geneticists and ecologists are debating right now. There may not be a setting that gives us everything we want. So what do we do?"*

Watch For

Revisit the opening poll. Has anyone changed their mind? Discuss what model evidence changed perspectives. Introduce the concept of irreversible decisions under uncertainty.

Slide 9: STEM Challenge

12 min

Say: *"The United Nations just appointed your team to design the rules for gene drive deployment. 600,000 lives hang in the balance — and so does the integrity of Earth's ecosystems. Design the framework."*

Watch For

Groups design governance frameworks including evidence thresholds, decision-making authority, monitoring requirements, and reversal strategies. Present and debate.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Gene Drive Efficiency
- **INTERNAL:** Mosquito Population, Malaria Transmission Rate, Wild-Type Gene Frequency, Ecological Dependence, Resistance Evolution, Geographic Spread Rate, Human Malaria Cases, Ecosystem Disruption Risk

Relationships

- **Gene Drive Efficiency to Wild-Type Gene Frequency:** NEGATIVE (-) — Higher Gene Drive Efficiency means more offspring inherit the drive construct instead of wild-type alleles, directly decreasing the frequency of natural genes in the population.
- **Wild-Type Gene Frequency to Mosquito Population:** POSITIVE (+) — As Wild-Type Gene Frequency drops (more mosquitoes carry the suppression drive), Mosquito Population declines because drive-carrying females have reduced fertility or the population is modified to prevent malaria transmission.
- **Mosquito Population to Malaria Transmission Rate:** POSITIVE (+) — Fewer mosquitoes means fewer opportunities for the Plasmodium parasite to be transmitted between human and mosquito hosts, directly reducing malaria transmission.
- **Mosquito Population to Ecosystem Disruption Risk:** NEGATIVE (-) — Declining Mosquito Population increases Ecosystem Disruption Risk as species that depend on mosquitoes for food lose a critical resource, triggering potential cascade effects through the food web.
- **Resistance Evolution to Gene Drive Efficiency:** NEGATIVE (-) — As Resistance Evolution increases in the mosquito population, the effective Gene Drive Efficiency decreases because resistant mosquitoes produce offspring that inherit wild-type alleles instead of the drive construct.

Key Discoveries

- Gene drives can theoretically spread a modified gene to an entire wild population within 10-20 generations, far faster than any natural evolutionary process — but this speed also means mistakes cannot be undone
- Resistance evolution is nearly inevitable — natural genetic variation ensures that some mosquitoes will carry mutations blocking the CRISPR mechanism, and selection will amplify these resistant genotypes
- The ecological consequences of mosquito elimination are uncertain but potentially significant — mosquitoes serve as food for thousands of species and as pollinators in some ecosystems
- Gene drives cannot be geographically contained — once released, the engineered gene will spread wherever the target species exists, making deployment an irreversible global decision

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Gene drives are like GMO crops — contained and controllable"	Gene drives are fundamentally different from GMO crops in one critical way: they are designed to spread. GMO crops contain modifications that follow normal Mendelian inheritance and stay in cultivated populations. Gene drives override Mendelian genetics and actively propagate through wild populations across generations and geographic boundaries. There is currently no proven technology to recall a gene drive once released.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If we eliminate mosquitoes, nothing bad will happen because they're just pests"	Mosquitoes play multiple ecological roles beyond being human pests. They are a primary food source for bats, birds, dragonflies, fish, and frogs. Mosquito larvae are a major component of freshwater food webs. Some mosquito species are pollinators. While individual ecological roles might be filled by other species, the cascading effects of removing an entire taxon from global ecosystems are genuinely unpredictable. Ecologists are genuinely divided on this question.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"CRISPR is perfectly precise and gene drives will work exactly as designed"	CRISPR is remarkably precise but not perfect. Off-target cuts occur, resistance mutations evolve, and molecular mechanisms can fail in ways not predicted by laboratory experiments. The model's Resistance Evolution component captures this reality — natural selection is relentless and creative. Every gene drive experiment in laboratory mosquito populations has observed resistance evolution within 10-25 generations. The question is whether the drive suppresses the population before resistance spreads, not whether resistance will emerge at all.